VWO Login Dashboard — Test Plan

Document control	Details
Product	VWO Login Dashboard
Source baseline	*Product Requirements Document: VWO Login Dashboard*
Product URL stated in PRD	`https://app.vwo.com/`
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> **Source-of-truth note:** This plan is based on the supplied VWO Login Dashboard PRD. It does not treat the separate VWO platform PRD, login API notes, or example credentials as requirements for this feature. Requirements that the PRD does not define are listed as open items and must be agreed before their tests are baselined.

1. Test Plan

This plan defines the approach, coverage, environments, entry and exit criteria, and release evidence for testing the VWO login dashboard. Testing is risk-based and covers the login journey from page arrival through authentication, recovery, and transition into the VWO dashboard.

2. Objective

Verify that users can securely and reliably access VWO through a clear, responsive, accessible login experience; that invalid or risky access is handled safely; and that the experience meets the PRD's measurable performance, availability, and business targets.

The plan is intended to:

- Confirm functional behavior against the login, validation, password recovery, session, MFA, SSO, and user-journey requirements.
- Identify regressions in usability, accessibility, security, privacy, and performance.
- Provide auditable test results and release risks for QA, Product, Engineering, Security, and Support stakeholders.

3. Product Context and Scope

The login dashboard is the entry point to VWO's experimentation, personalization, and analytics platform. Its users include digital marketers, product managers, UX designers, developers, CRO specialists, analysts, and enterprise teams.

3.1 In scope

1. Login page availability, branding, layout, and loading behavior.
2. Email/password authentication, input validation, error feedback, and loading states.
3. Remember Me behavior and secure session creation, persistence, expiration, and termination.
4. Forgot-password and email-based password recovery, including token behavior once the policy is supplied.
5. Optional 2FA, enterprise SSO (including SAML/OAuth as applicable), and social identity providers only where enabled for the test release.
6. Successful transition to the correct VWO dashboard and handling of login success/failure analytics.
7. New-user registration/free-trial link and support-assistance paths.
8. Responsive behavior, light/dark theme, keyboard access, screen-reader support, high contrast, and WCAG 2.1 AA verification.
9. Security and privacy controls, including HTTPS, rate limiting, session protections, GDPR/CCPA handling, and enterprise audit requirements.
10. Performance and reliability checks against the explicit PRD targets.

3.2 Out of scope

- Functional testing of the VWO experimentation, personalization, analytics, or other authenticated product modules, except verifying the login-to-dashboard handoff.
- Full onboarding, subscription, billing, or trial-to-paid workflows beyond confirming the registration entry point.
- Implementation of biometric authentication, adaptive authentication, PWA behavior, or future login personalization unless explicitly included in the release scope.
- Penetration testing against production or third-party identity providers. Security testing must use approved non-production environments and authorization.
- Verifying a specific API schema, UI selector, exact error copy, or password policy not specified by the PRD.

4. Requirements Traceability

The identifiers below are QA tracking IDs assigned for this plan; the PRD itself does not provide requirement IDs.

QA ID	PRD requirement	Planned verification
LGN-01	Email/password authentication	Valid, invalid, inactive, and malformed account scenarios; authentication outcome and dashboard handoff
LGN-02	Secure session management with configurable timeout	Session creation, Remember Me, expiry, logout/termination, and protected-page access

QA ID	PRD requirement	Planned verification
LGN-03	Optional 2FA	Challenge, valid/invalid/expired verification, and recovery behavior when 2FA is enabled
LGN-04	Enterprise SSO; SAML/OAuth and other supported protocols	Redirect, successful/failed identity-provider response, cancellation, and return-to-app behavior when configured
LGN-05	Real-time validation on blur and email format verification	Valid/invalid email, blur timing, focus, and clear recovery from validation errors
LGN-06	Password strength feedback and password requirements	Feedback and enforcement against the approved password policy
LGN-07	Clear, actionable authentication error handling	Invalid credentials, service failure, throttling, and non-disclosing error behavior
LGN-08	Forgot password and email-based recovery with secure token generation	Request, delivery, valid/invalid/expired/reused token, completion, and follow-up login
LGN-09	Responsive, mobile-optimized, touch-friendly interface	Supported viewport/device matrix and orientation/zoom behavior
LGN-10	Autofocus, clickable labels, and loading states	Initial focus, label activation, visible/busy state, and duplicate-submit prevention
LGN-11	Screen-reader support, ARIA, keyboard navigation, high contrast	WCAG 2.1 AA-oriented keyboard, assistive-technology, focus, contrast, and semantics checks
LGN-12	VWO branding and light/dark theme	Logo, typography, visual consistency, and theme rendering/state
LGN-13	Encryption, secure password storage, secure session tokens, HTTPS	Transport/security configuration review and approved security verification; password storage requires backend evidence
LGN-14	GDPR/CCPA and enterprise policy/audit requirements	Data minimization, privacy-policy alignment, retention/deletion evidence, and audit events against approved policy
LGN-15	Rate limiting and brute-force protection	Threshold, response, recovery, and monitoring behavior against agreed security policy

QA ID	PRD requirement	Planned verification
LGN-16	Login page loads within 2 seconds on standard connections	Repeatable page-load measurement in a defined standard-connection test environment
LGN-17	99.9% availability and thousands of concurrent login attempts	Availability evidence and authorized load testing at an agreed concurrency profile
LGN-18	CDN, asset optimization, and global performance	Asset delivery, caching, and regional measurements in approved environments
LGN-19	Seamless dashboard transition and recent activity/context	Correct destination, user/account context, and session restoration per product behavior
LGN-20	Success/failure analytics and support integration	Event correctness, no credential leakage, and functional support entry point
LGN-21	Registration/free-trial entry and new-user discovery	Correct link/destination and basic navigation from the login page
LGN-22	Success metrics: 95% login success, 90% satisfaction, 20% fewer support tickets	Define instrumentation/population and baseline; report measurable outcomes without treating a single test as proof of the KPI

5. Inclusions and Test Scenarios

Priorities: **P0** = release-blocking core/security risk; **P1** = high-value required coverage; **P2** = supporting, compatibility, or metric coverage. Phase 1/2/3 follow the development phases in the PRD; they are not claims that a capability is currently implemented.

Scenario ID	Scenario	Expected result	Priority / phase
VWO-LGN-001	Open login page over HTTPS on a supported environment	Page renders without certificate/security warnings and all required controls are available	P0 / 1
VWO-LGN-002	Sign in with an active account and valid credentials	Authentication succeeds once, a secure session is established, and the user reaches the correct dashboard/account	P0 / 1
VWO-LGN-003	Submit incorrect password or unknown email	Access is denied; feedback is actionable but does not disclose whether an account exists	P0 / 1

Scenario ID	Scenario	Expected result	Priority / phase
VWO-LGN-004	Submit with empty, malformed, boundary-length, or whitespace-containing email/password	Validation matches approved rules, identifies recoverable input errors, and does not create a session	P0 / 1
VWO-LGN-005	Trigger email validation on blur and use the clickable field label	Validation timing and label activation work without disrupting keyboard/screen-reader use	P1 / 1
VWO-LGN-006	Test password-strength feedback and password policy boundaries	Feedback and server-side enforcement agree with the approved policy; password is never exposed in logs or UI	P1 / 1
VWO-LGN-007	Submit repeatedly or double-click while authentication is pending	A clear loading/busy state appears; accidental duplicate submissions do not create inconsistent sessions	P1 / 1
VWO-LGN-008	Use Forgot Password with registered, unregistered, malformed, and blank addresses	Flow follows the agreed privacy-safe response; only eligible users receive recovery instructions	P0 / 1
VWO-LGN-009	Complete reset with valid, invalid, expired, reused, and tampered recovery tokens	Only a valid, unexpired, unused token accepted under policy can complete reset; token lifecycle is enforced	P0 / 1
VWO-LGN-010	Sign in using the newly reset password and attempt the prior password	New credential works; prior credential is invalidated as required by approved policy	P0 / 1
VWO-LGN-011	Toggle Remember Me on and off across browser restart	Persistence and expiry match the documented choice; sensitive credentials are not stored in plaintext	P1 / 1
VWO-LGN-012	Let a session expire, log out, or revisit a protected URL after termination	Protected access is denied and the user is directed to authentication without exposing private content	P0 / 1
VWO-LGN-013	Complete, fail, cancel, and retry 2FA when enabled	Only a valid challenge grants access; failures/expiry/recovery follow configured policy	P0 / 1
VWO-LGN-014	Complete, cancel, fail, and retry enterprise SSO for configured SAML/OAuth providers	State/return URL is handled safely, errors are recoverable, and only the expected identity is authenticated	P0 / 3
VWO-LGN-015	Use configured Google/Microsoft or other social login option	Provider redirect/callback, cancellation, denied consent, and error cases are handled safely	P1 / when enabled

Scenario ID	Scenario	Expected result	Priority / phase
VWO-LGN-016	Navigate all login and recovery controls using keyboard only	Logical focus order, visible focus, operable controls, and no keyboard trap	P0 / 2
VWO-LGN-017	Verify screen-reader names, roles, states, validation messages, and live status	Controls and dynamic feedback are announced accurately and meaningfully	P0 / 2
VWO-LGN-018	Verify contrast/high-contrast mode, zoom, and text resizing	Content and controls remain perceivable and operable against the approved WCAG 2.1 AA criteria	P1 / 2
VWO-LGN-019	Render supported desktop/mobile widths, orientations, and touch targets	No clipped/overlapping controls; content reflows and touch controls remain usable	P0 / 2
VWO-LGN-020	Switch light/dark theme and reload/navigate	Theme assets and contrast remain correct; persistence matches the specified behavior	P1 / 2
VWO-LGN-021	Verify VWO logo, colors, typography, links, and page states	Login dashboard is consistent with the approved VWO design system and current approved UI	P1 / 1
VWO-LGN-022	Verify login success/failure analytics and support handoff	Correct events and safe metadata are emitted; credentials, tokens, and sensitive personal data are not included	P1 / 3
VWO-LGN-023	Follow the free-trial/registration and support links	Each link reaches the approved destination and preserves a usable return path when defined	P2 / 1
VWO-LGN-024	Measure login page load on standard connection	Agreed page-load metric is within the PRD's 2-second target; assets are served as designed	P0 / 1
VWO-LGN-025	Exercise concurrent authentication and availability under approved load	Service remains stable and secure at agreed thousands-of-users profile; latency/error thresholds are agreed before execution	P0 / 3
VWO-LGN-026	Review login service availability and regional/CDN behavior	Evidence supports the 99.9% availability target and expected global delivery; failures are observable	P1 / 3
VWO-LGN-027	Check privacy and enterprise audit behavior	Data handling, consent/notice, retention, and audit trail align with approved GDPR/CCPA and enterprise policies	P0 / 3

Scenario ID	Scenario	Expected result	Priority / phase
VWO-LGN-028	Verify password and session security controls	HTTPS, secure token handling, no credential leakage, and approved password-storage controls are demonstrated	P0 / 1
VWO-LGN-029	Exercise rate limits and recovery after repeated failures	Throttling/mitigation follows approved policy, does not enable account enumeration, and legitimate users can recover	P0 / 1
VWO-LGN-030	Validate 95% login success and user/support KPIs	Instrumentation, numerator/denominator, sampling window, and baseline are agreed; results are reported with context	P2 / ongoing

5.1 Critical end-to-end flows

Returning user: Open login page → enter valid credentials → submit → complete 2FA/SSO if configured → reach the expected VWO dashboard → verify account context → terminate session → confirm protected content is no longer accessible.

Recovery: Open Forgot Password → submit account identifier → receive privacy-safe response → use valid reset link/token → set a compliant password → sign in with the new password → confirm token reuse and the prior password are rejected as policy requires.

Accessible mobile login: Open supported mobile viewport → navigate and submit with keyboard/assistive technology → receive accessible validation/loading feedback → complete authentication → verify no content or controls are obscured.

6. Test Environments

The PRD identifies `https://app.vwo.com/` but does not identify a QA or staging URL. Testing must use a designated non-production environment for functional, security, and load tests. Production smoke checks, if required, must be explicitly authorized, read-only, and use approved accounts.

Environment dimension	Planned coverage / decision needed
Application	QA/staging URL and release/build identifier: **TBD**; production URL only for approved low-risk checks
Identity/data	Seeded synthetic users: active, inactive, MFA-enabled, SSO-enabled, and recovery-capable accounts; no real customer credentials
Browsers	Current supported Chrome, Edge, Firefox, and Safari versions: **confirm support matrix**
Devices	Desktop and representative iOS/Android phones/tablets: **confirm device/OS support matrix**

Environment dimension	Planned coverage / decision needed
Network	Standard connection for the 2-second target; throttled/mobile and regional/CDN profiles for resilience checks
Accessibility	Keyboard-only, screen-reader/browser combinations, zoom/high contrast, and automated accessibility checks
Load	Isolated authorized environment, controlled test accounts, agreed concurrent-user profile, monitoring, and stop conditions
Security	Approved test scope, test accounts, logging access, rate-limit policy, and security sign-off

7. Test Strategy

1. **Requirement analysis:** Review PRD, designs, security/privacy policies, supported environment matrix, and release scope. Resolve open items before assigning final pass/fail criteria.
2. **Test design:** Use equivalence partitioning and boundary analysis for inputs; decision tables for Remember Me, MFA, SSO, and account states; state-transition testing for login/recovery/session/token flows; use-case testing for end-to-end journeys; and risk-based exploratory testing.
3. **Shift-left reviews:** Review form validation, error copy, accessibility semantics, security controls, telemetry, and API contracts as they are defined, before UI freeze.
4. **Build verification:** Run smoke tests for page availability, successful/failed authentication, reset entry point, and dashboard transition. Reject the build for a P0 smoke failure.
5. **Functional and integration testing:** Execute UI and service-level tests in QA/staging. Verify provider integrations only with approved sandbox tenants.
6. **Cross-browser/device and accessibility:** Run the agreed matrix. Combine automated checks with manual keyboard and assistive-technology review; automated scans alone do not establish WCAG conformance.
7. **Security and privacy:** Verify controls against approved policy and conduct authorized security testing. Do not send secrets to logs or execute destructive tests on production.
8. **Performance/reliability:** Measure the login-page 2-second target under a recorded standard network profile. Run concurrency, regional, and availability tests in an approved environment with agreed thresholds and monitoring.
9. **Regression and retest:** Retest fixes and run login/recovery/session regression on every release candidate; expand to the full matrix for changes to shared authentication, UI, or identity integrations.
10. **Exploratory testing:** Investigate browser navigation, refresh/back behavior, network interruption, slow responses, repeated submissions, localization/long text if supported, and error recovery.

7.1 Automation approach

Use the repository's Playwright JavaScript/TypeScript setup if the team elects to automate these scenarios. Automate stable, high-value paths first: successful/failed login, required-field validation, recovery entry, dashboard handoff, Remember Me, session expiry, and key viewport checks. Use test IDs or accessible role/name locators agreed with Engineering; avoid brittle selectors. Keep credentials and tokens in approved secret storage/environment variables, never in source control or test reports. Exact selectors, fixtures, and CI integration are implementation prerequisites, not assumptions of this plan.

8. Test Schedule

Dates, sprint length, staffing, and release milestones were not supplied. The following is a dependency-based sequence; the QA lead should add calendar dates after scope and environment are confirmed.

Phase	Activities	Exit evidence
0. Readiness	Resolve open items, confirm non-production environment, requirements, data, and approvals	Reviewed requirements and ready-to-test checklist
1. Core authentication	Smoke, login, validation, password recovery, session, baseline security, and page-load checks	Core scenario results and P0 defect status
2. UX and accessibility	Responsive/mobile, themes, keyboard, screen reader, contrast, and usability coverage	Compatibility and accessibility results
3. Enterprise and NFR	MFA, SSO/social providers as released, privacy/audit, rate limits, concurrency, availability, and regional checks	Security/performance evidence and residual risks
4. Regression and closure	Fix verification, regression, KPI/instrumentation review, summary, and sign-off	Final test summary and release recommendation

9. Test Deliverables

- This reviewed test plan.
- Requirement-to-test traceability and scenario inventory.
- Approved test data/environment matrix and automation coverage, where applicable.
- Execution results with evidence and build/environment identifiers.
- Defect records with reproducible steps and severity/priority.
- Security, privacy, performance, accessibility, and compatibility evidence for the scope executed.
- Test summary, residual risks, open decisions, and release recommendation.

10. Defect Reporting Procedure

Record defects in the project-approved tracking system (**tool/project: TBD**). Each report should include:

- Unique ID, concise title, affected requirement/scenario, severity, priority, and build/environment.
- Preconditions, exact reproduction steps, expected and actual results, reproducibility, and timestamps.
- Redacted screenshots, sanitized console/network traces, and relevant correlation IDs.
- Impacted browser/device/account state and whether the issue blocks a core journey.
- No raw passwords, authentication tokens, reset links, session cookies, or unnecessary personal data.

Triage P0 security/authentication/data exposure issues immediately with QA, Engineering, Product, and Security. Retest the fix in the original environment and run focused regression before closure. Escalation contacts and service-level targets are **TBD**.

11. Entry and Exit Criteria

11.1 Requirement analysis

Entry: PRD and release scope are available to QA. **Exit:** Requirements are reviewed; testable acceptance criteria, traceability, unresolved decisions, and responsible owners are recorded.

11.2 Test design and readiness

Entry: Release scope and primary flows are identified. **Exit:** Scenarios are reviewed; supported browser/device matrix, non-production URL, test accounts, security authorization, and applicable pass/fail thresholds are confirmed.

11.3 Test execution

Entry:

- Build is deployed and accessible in the approved test environment.
- Smoke tests pass; test accounts/data and required identity-provider sandboxes are available.
- Designs, validation/error copy, password/session/rate-limit policies, and acceptance criteria needed for the planned tests are approved.
- Load/security test authorization and operating limits are documented before those tests begin.

Exit:

- Planned in-scope P0/P1 scenarios have results; failures and blocked tests have owners and disposition.
- All P0 tests pass, or any exception is explicitly accepted by accountable Product, Engineering, and Security owners.
- No unresolved critical/high defects remain unless formally risk-accepted.
- Fixes are retested and required regression has passed.
- Performance, accessibility, security/privacy, and compatibility evidence is attached for the executed scope.
- Residual risks, open requirements, and KPI measurement limitations are reported.

11.4 Test closure

Entry: Execution results and defect status are available. **Exit:** Test summary and recommendation are delivered; evidence is archived according to policy; stakeholders have recorded approval or accepted risk.

12. Test Closure and Reporting

The summary report should show planned/executed/passed/failed/blocked tests by priority and requirement, open defects by severity, test environment/build, and outstanding risks. Report performance measurements with network profile and percentile/aggregation method. Report login success, satisfaction, and support-volume KPIs only after the metric definitions, measurement window, sample, and baseline are agreed.

13. Tools

Purpose	Proposed tool / status
UI end-to-end automation	Playwright (JavaScript/TypeScript), subject to repository and team approval

Purpose	Proposed tool / status
Defect management	Project-approved issue tracker: **TBD**
Browser/device coverage	Supported browser/device farm or approved local matrix: **TBD**
Accessibility	Automated accessibility scanner plus manual keyboard/screen-reader checks
Performance/load	Approved monitoring and load-testing tools: **TBD**
Evidence and reporting	Approved document, screenshot, and CI artifact storage

14. Risks and Mitigations

Risk / dependency	Impact	Mitigation
PRD does not define exact validation/error copy, input limits, or password rules	Inconsistent or subjective expected results	Product/Security approve field rules and message behavior before baseline
No QA URL, build, browser matrix, or test-account details are supplied	Tests may be unsafe, blocked, or non-representative	Provision non-production environment, synthetic accounts, and supported matrix before execution
Session timeout, Remember Me persistence, token lifetime, and logout behavior are unspecified	Session checks cannot have objective pass/fail outcomes	Security/Product publish session and recovery-token policies
MFA, SSO, and social login scope depends on release phase/provider setup	Integration scenarios may be unavailable	Mark each capability in/out per release; provide provider sandbox configuration and test identities
“Thousands” of concurrent users and 99.9% uptime lack workload/measurement definitions	Load/availability result may be misleading	Agree concurrency mix, duration, latency/error SLOs, observation period, and ownership before testing
GDPR/CCPA, enterprise audits, and telemetry details are high-level	Compliance coverage cannot be concluded from UI tests alone	Obtain approved data-flow, retention, consent, audit, and legal/security requirements

Risk / dependency	Impact	Mitigation
Production-directed security/load tests could affect users or data	Service disruption or privacy incident	Restrict invasive testing to authorized isolated environments; document approvals and stop conditions
Identity provider, email delivery, CDN, or network instability	False failures and delayed verification	Monitor dependencies, capture correlation IDs, and distinguish product defects from external outages

15. Open Items and Assumptions

The items below are not defined by the supplied PRD and require confirmation. They are not silently assumed as product behavior.

1. Which capabilities are included in this release: 2FA, SAML/OAuth SSO, social login, support integration, and account recovery options?
2. What are the exact field constraints, password complexity rules, validation timing, and user-facing error messages?
3. What is the approved account-enumeration policy for login and password-reset responses?
4. What are the session timeout, Remember Me duration, token expiry/reuse, logout, and concurrent-session rules?
5. Which browsers, OS versions, devices, locales, and assistive technologies are officially supported?
6. Which QA/staging URL, seeded accounts, identity-provider sandbox, email test inbox, and build pipeline are available?
7. What standard-network profile and page-load measurement definition determine the 2-second target?
8. What workload, duration, geographic regions, and latency/error thresholds define “thousands” of concurrent attempts and 99.9% availability?
9. Which privacy notices, consent rules, retention/deletion periods, audit events, and data residency rules apply?
10. What event names/fields define login success/failure analytics, and how are the 95% success, 90% satisfaction, and 20% support-ticket reduction KPIs measured?
11. What issue tracker, severity definitions, escalation path, and approval roles should be used?

16. Approvals

Role	Name	Approval / date
Product Owner	TBD	Pending
Engineering Lead	TBD	Pending
QA Lead	TBD	Pending

Role	Name	Approval / date
Security / Privacy	TBD	Pending

Testing beyond the agreed scope or against production may begin only with the relevant explicit approval.